Module 11: The Future of AI Engineering

Synthesis, trends, and personal practice

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Primary sources: Osmani, Chapter 11; Thomas & Hunt, Essence of Good Design.

Learning Outcomes

- Evaluate emerging AI-assisted development trends.
- Synthesize course concepts into a framework.
- Propose improvements to AI-assisted workflows.
- Formulate a personal engineering approach to AI use.

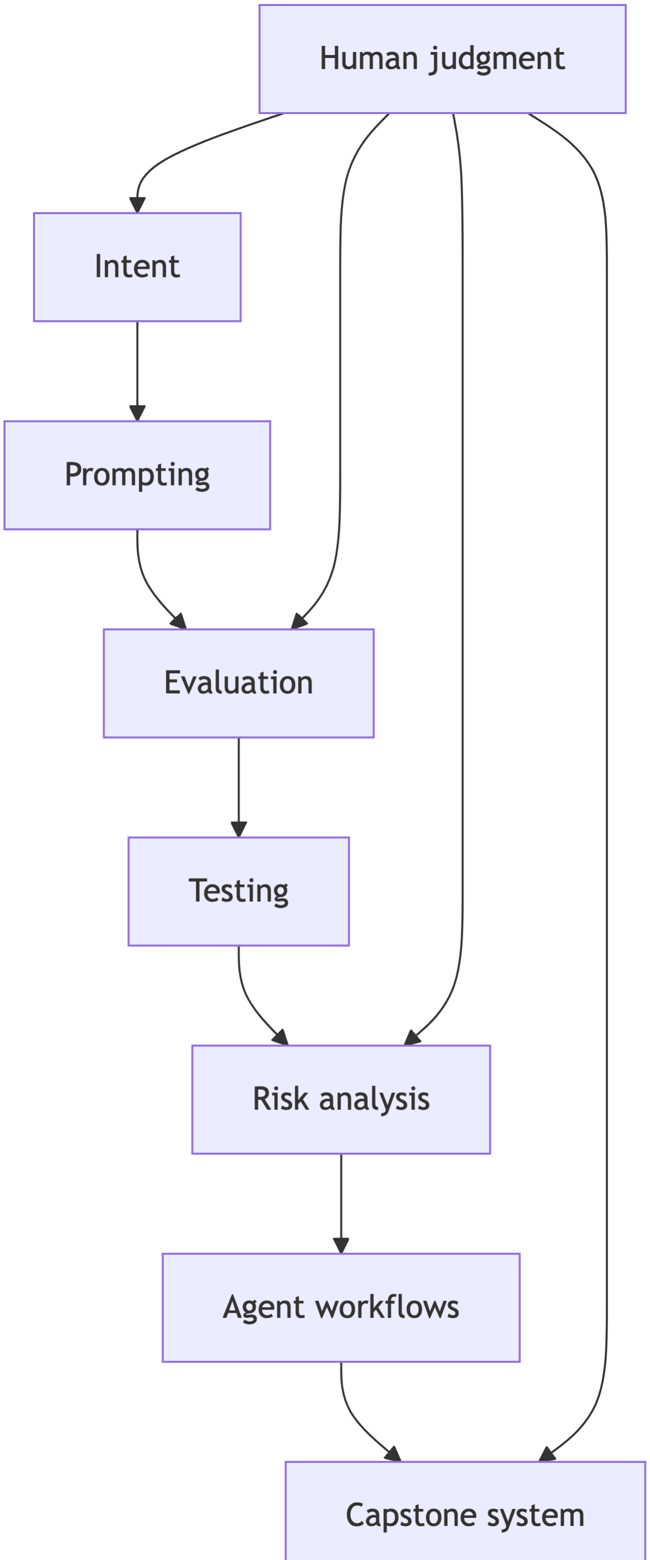
Beyond Code Generation

AI assistance is expanding into:

- Test generation.
- Debugging.
- Maintenance.
- Design exploration.
- Project coordination.
- Agent-driven task execution.

The engineering problem remains: how do we trust the result?

Course Concept Map



Future Skill Shift

Important skills move toward:

- Problem framing.
- Requirements clarification.
- Architecture judgment.
- Test design.
- Review and verification.
- Workflow design.
- Ethical decision-making.

Essence of Good Design

Thomas and Hunt define good design around ease of change.

AI-era interpretation:

- Good systems absorb generated changes safely.
- Good boundaries make AI suggestions reviewable.
- Good tests make iteration cheap.
- Good documentation makes context reusable.

Personal AI Workflow

A durable workflow should answer:

- When do I ask AI for help?
- How do I scope the request?
- What evidence must I collect?
- What do I never delegate?
- How do I improve after each interaction?

Course Framework

The course has built a progression:

- Intent.
- Prompting.
- Evaluation.
- Human judgment.
- Testing.
- Production readiness.
- Systems thinking.
- Security.
- Ethics.
- Agents.
- Capstone integration.

Milestone 2 Connection

Your partial implementation should prove the final system is feasible.

Evidence should include:

- Meaningful implementation.
- Component integration.
- Example workflow run.
- Evaluation strategy.
- Risks and limitations.
- Peer review and AI notes.

Improving the Workflow

Ask your team:

- Which prompts produced useful output?
- Which AI suggestions created rework?
- Which tests found real issues?
- Which review practices improved quality?
- Which risks are still unmanaged?

Practice Activity

Create a personal AI-assisted SWE checklist with three sections:

- Before prompting.
- Before accepting output.
- Before submitting or merging.

Tie each item to course evidence.

Takeaways

- The future is not just faster coding.
- The future is better orchestration, verification, and judgment.
- Your professional advantage is disciplined AI use.